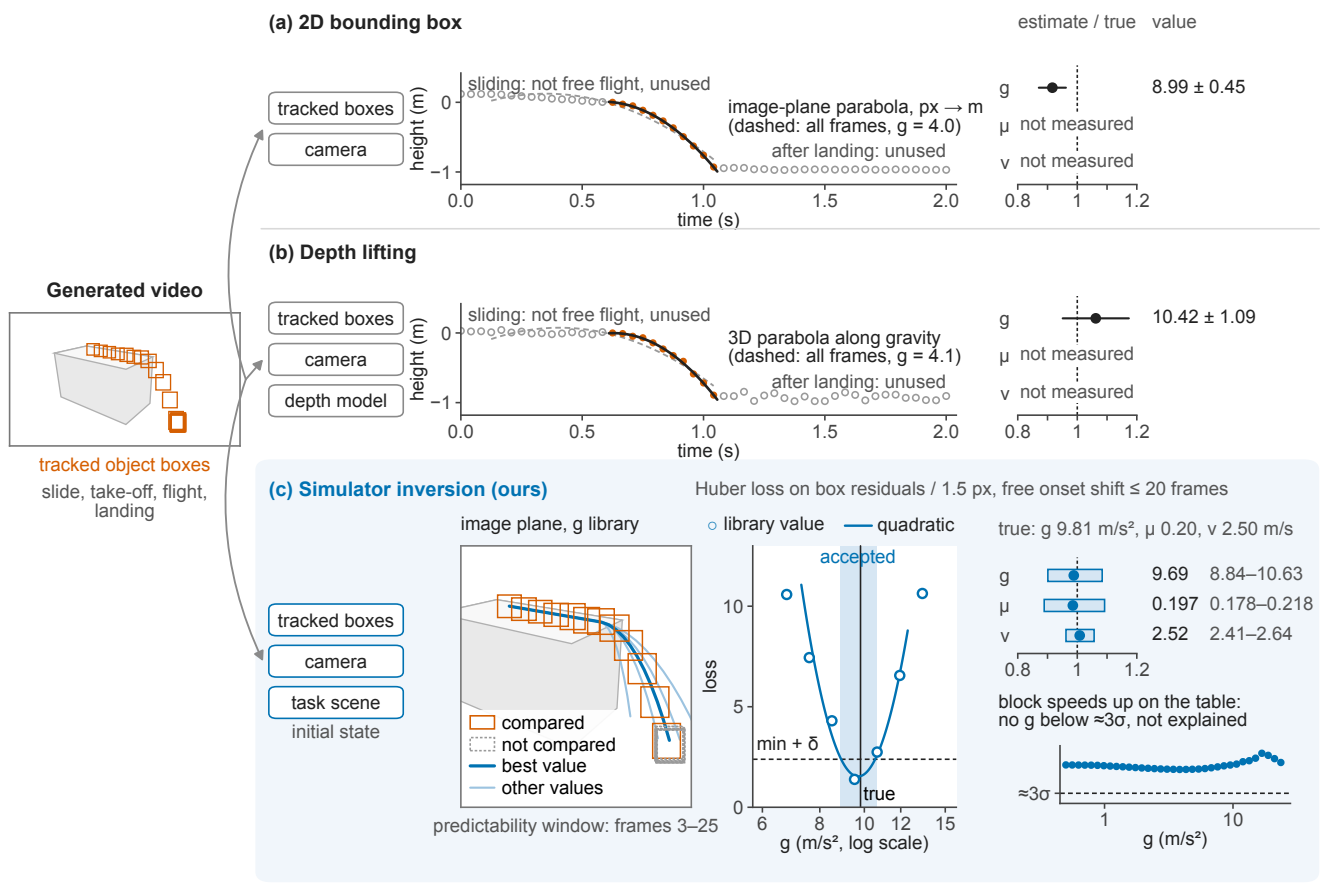

(a) 2D bounding box
estimate / true value
tracked boxes
camera
sliding: not free flight, unused
height (m)
image-plane parabola, px → m
(dashed: all frames, g = 4.0)
after landing: unused
time (s)
g
8.99 ± 0.45
μ not measured
v not measured
(b) Depth lifting
tracked boxes
camera
depth model
sliding: not free flight, unused
height (m)
3D parabola along gravity
(dashed: all frames, g = 4.1)
after landing: unused
time (s)
g
10.42 ± 1.09
μ not measured
v not measured
Generated video
tracked object boxes
slide, take-off, flight, landing
(c) Simulator inversion (ours)
Huber loss on box residuals / 1.5 px, free onset shift ≤ 20 frames
tracked boxes
camera
task scene
initial state
image plane, g library
compared
not compared
best value
other values
predictability window: frames 3–25
library value
quadratic
accepted
loss
min + δ
true
g (m/s², log scale)
true: g 9.81 m/s², μ 0.20, v 2.50 m/s
g 9.69 8.84–10.63
μ 0.197 0.178–0.218
v 2.52 2.41–2.64
block speeds up on the table:
no g below ≈3σ, not explained
≈3σ
g (m/s²)